

Product Specification: Architectural Scenario Builder ("Architectural What-If Simulator")

1. Executive Summary & Purpose

The **Architectural Scenario Builder** is a dedicated enterprise architecture simulation tool designed to answer "What-If" structural questions across all four TOGAF / ArchiMate layers: **Business, Information, Application, and Technology**.

Unlike financial P&L simulators, this tool evaluates the structural changes, dependency disruptions, operational risks, and cross-layer ripple effects when architectural components are replaced, added, removed, or refactored.

2. Target Persona & Key Use Cases

Target Personas

- **Enterprise & Solution Architects:** Modeling target architectures, technology stack modernizations, and COTS evaluations.
- **CTOs & Tech Leads:** Assessing technical debt, migration impact, and architectural risk profiles.
- **Domain & Business Architects:** Understanding how IT/App layer modifications impact business processes, capabilities, and data flows.

Core Use Cases

1. **Component Refactoring:** *"What if we replace direct database queries across 12 microservices with a unified GraphQL API layer?"*
2. **COTS / SaaS Replacement:** *"What if we replace our custom-built CRM module with Salesforce SaaS?"*
3. **Infrastructure Modernization:** *"What if we migrate on-prem SQL servers to cloud-native serverless databases?"*
4. **Decommissioning & M&A Consolidation:** *"What if we decommission legacy Application X and reroute its business functions to Application Y?"*
5. **Open-Ended / Complex Refactoring:** Custom plain-language scenarios combining multiple structural shifts across layers.

3. UI/UX Architecture & Layout

The user interface follows a streamlined **split-view, node-graph-centric canvas pattern** focused entirely on topological clarity, direct canvas interaction, and multi-layered visual

feedback.

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| TOP BAR: Scenario Selector | Baseline vs. To-Be Toggle | Run Simulation Button | Export Spec
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| LEFT PANEL          | MAIN CANVAS (Center View)                                | RIGHT PANEL
|
|                    |                    |
|
| 1. Target Entity | [TOGAF Layered Topology Graph]                            | Cross-Layer Impact
|
|   Picker         |                    | Inspector
|
|   (Canvas Click)|   (B) Business Layer: [Procurement]                        |
|
|                    |                    | * Affected Entities
|
| 2. Transformation|   (I) Info Layer:      [Order Schema]                      |
|
|   Config         |                    | * Dependency Risk
Score|
|   - Structured  |   (A) App Layer:      [Legacy ERP] -> (API Adapter) | * Structural Metrics
|
|     Actions     |                    |
|
|   - Custom Free-|   (T) Tech Layer:     [On-Prem Oracle DB]                |
|
|   Text Prompt   |                    |
|
|                    | (Color-coded Heatmap: Green/Yellow/Red/Purple)           |
|
| 3. Layer Filters |                    |
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4. Detailed Panel Specifications

A. Left Panel: Scenario Configuration & Control

- **Target Node Selector (Primary Interaction: Canvas Click):**
 - Direct click-to-select on any node within the interactive canvas to designate the anchor component for the scenario.
 - Secondary search/filter dropdown for quick node lookup in dense architectures.
- **Transformation Action Selector:**
 - **Structured Transformation Actions:**
 - **Replace:** Swap selected component with a COTS package or new technology.
 - **Decouple / Reroute:** Introduce an abstraction layer (e.g., replace DB coupling

with REST/GraphQL API or message broker).

- **Decommission / Remove:** Sunset an application or infrastructure component.
- **Consolidate:** Merge multiple components into one unified platform.
- **Custom Natural Language / Free-Text Scenario Prompt:**
 - An open-ended text entry field allowing architects to input arbitrary or hybrid scenarios in plain language (e.g., *"Split the monolithic order management system into 3 microservices and route all event logs through Apache Kafka"*).
- **Action Parameters & Constraints:**
 - Proposed Alternative / Target COTS platform (e.g., Salesforce API, AWS Aurora Postgres, Kafka).
 - SLA / Compliance constraints (e.g., *Zero downtime required, GDPR data boundary*).
- **Layer Toggles:** Filter canvas display by layer (Business, Information, Application, Technology).

B. Center Canvas: Interactive Architecture Impact Map

- **Layered Swimlane View:**
 - Horizontal or vertical swimlanes corresponding to **Business, Information, Application**, and **Technology** layers.
 - Explicit dependency links connecting nodes across layers.
- **Visual Delta Encoding:**
 - **Red / Amber Nodes:** Directly and indirectly broken dependencies or high-risk impact zones.
 - **Purple / Dotted Nodes:** New components introduced in the "To-Be" scenario.
 - **Grey / Strikethrough Nodes:** Decommissioned or replaced legacy components.
 - **Green Nodes:** Unaffected baseline components.
- **Interactive Canvas Features:**
 - **Click-to-Select Anchor Node:** Single-clicking any canvas node sets it as the target entity in the Scenario Panel.
 - **Ripple Effect Animation:** Triggering a simulation animates glowing wave pulses radiating from the target component outward to dependent nodes across all layers.
 - **Side-by-Side Dual Canvas Mode:** Compare "As-Is" architecture on the left with "To-Be" architecture on the right.
 - **In-Canvas Context Menu:** Right-click any node to "Simulate Removal", "Replace with API", or "Inspect Dependencies".

C. Right Panel: Cross-Layer Impact Inspector

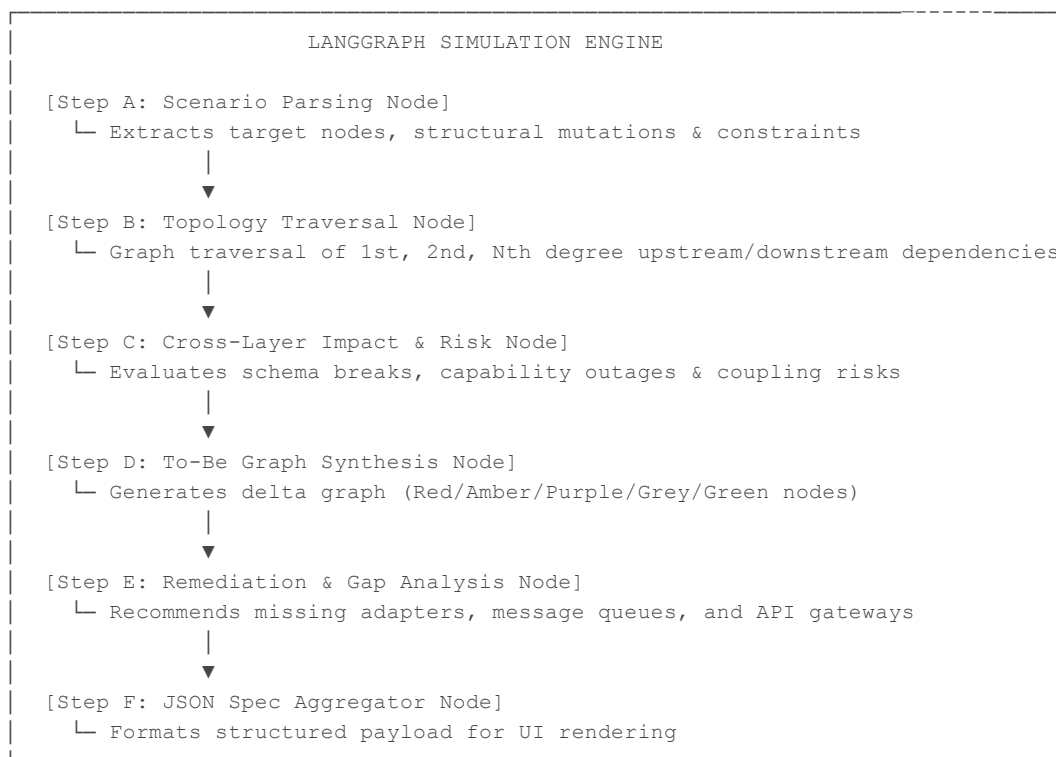
- **Impact Heatmap & Structural Metrics:**
 - **Structural Risk Score (0-100):** Measure of dynamic coupling and potential integration failure.
 - **Data Integrity & Flow Risk:** Identifies data schemas, contracts, or interfaces broken by the scenario.
 - **Business Capability Vulnerability:** Highlights which critical business functions (e.g., "Order Fulfillment") are impacted.

- **Detailed Breakdown Tabs:**
 1. **Directly Impacted:** Nodes directly linked to the changed component.
 2. **Indirect / Ripple Impacted:** Downstream systems, APIs, reports, and workflows affected 2+ steps away.
 3. **Gap Analysis:** Missing adapters, security protocols, or interfaces required to realize the target architecture.

5. Core Multi-Step LangGraph Simulation Workflow

Because a single-prompt LLM call is insufficient for rigorous architectural impact analysis, the simulation engine utilizes a multi-step **LangGraph** orchestration flow. The execution graph processes the transformation through dedicated, stateful analysis nodes:

[1. User Input & Canvas Click]



[2. Render Ripple Heatmap, To-Be Graph & Cross-Layer Inspector in UI]

Detailed LangGraph Node Specifications:

1. **Step A: Scenario & Intent Parsing Node:**
 - **Input:** User structured inputs (Node ID, Action) and/or natural language custom text prompt.

- **Operation:** Parses exact entities, action intent (replace, decouple, decommission, consolidate), and architectural constraints.
- **Output:** Normalized mutation specification object.
- 2. **Step B: Graph Dependency Traversal Node:**
 - **Input:** Topology graph model + target component ID.
 - **Operation:** Executes multi-degree depth-first and breadth-first search across TOGAF layers (Technology ↔ Application ↔ Information ↔ Business).
 - **Output:** Active subgraph containing directly and transitively connected components.
- 3. **Step C: Cross-Layer Impact & Risk Evaluation Node:**
 - **Input:** Active subgraph + mutation spec.
 - **Operation:** Analyzes protocol mismatches, broken data schemas, lost capabilities, and runtime coupling points.
 - **Output:** Risk scores, broken interface list, and impacted business processes.
- 4. **Step D: To-Be Graph Synthesis Node:**
 - **Input:** Baseline graph + structural mutations.
 - **Operation:** Constructs the "To-Be" topology graph, adding new COTS/API components (Purple), marking removed nodes (Grey), flagging broken dependencies (Amber/Red), and keeping unchanged nodes (Green).
 - **Output:** Complete To-Be state graph payload.
- 5. **Step E: Architectural Remediation & Gap Analysis Node:**
 - **Input:** Impact assessment + To-Be graph.
 - **Operation:** Identifies missing architectural middleware (e.g., event buses, security proxies, database adapters) needed to repair red/amber dependency breaks.
 - **Output:** Remediation action plan and gap list.
- 6. **Step F: JSON Spec Aggregator Node:**
 - **Input:** Outputs from Steps A through E.
 - **Operation:** Synthesizes final JSON response conforming to the frontend architectural_scenario state schema for real-time visualization.

6. Frontend Component Tree (architectural-scenario-viewer.tsx)

- ArchitecturalScenarioViewer (Main Container)
 - ScenarioHeaderBar (Baseline vs. To-Be toggle, preset dropdown, simulation status & trigger)
 - MainWorkspace (Flex layout)
 - ScenarioConfigSidebar (Left panel: click target status, structured actions, custom prompt text box)
 - ArchitectureCanvas (React Flow / Cytoscape graph renderer)
 - LayerSwimlanes (Business, Info, App, Tech visual bands)
 - ArchitectureNode (Custom graph nodes with direct click handlers & health indicators)

- DependencyEdge (Custom directed arrows with data flow badges)
- ImpactInspectorSidebar (Right panel: structural risk breakdown, affected capabilities, gap analysis)